

BEAVERLODGE, AB, Canada

WMO: 712300

Lat: 55.1967N

Lon: 119.3964W

Elev: 2444

StdP: 13.44

Time Zone: -7.00 (NAM)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.1	-17.5	-31.9	1.0	-23.5	-24.8	1.5	-17.2	31.0	38.2	27.4	37.9	2.9	40	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	80.3	59.0	77.0	57.7	73.9	56.6	61.9	74.4	60.0	72.3	58.4	70.0	8.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.5	77.2	65.4	55.3	71.3	63.3	53.5	66.6	62.3	28.9	74.6	27.5	73.1	26.4	69.9	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.2	21.5	18.5	DB	-32.2	86.4	7.0	3.3	-37.2	88.8	-41.3	90.7	-45.2	92.6	-50.3	95.0	(4)
(5)			WB	-32.4	65.0	6.9	2.5	-37.4	66.8	-41.4	68.3	-45.3	69.7	-50.3	71.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.9	12.8	17.3	23.5	37.7	49.7	56.6	60.4	58.4	49.9	38.4	22.7	14.5		
	DBStd	20.49	17.33	14.20	13.47	9.03	7.90	5.51	5.25	5.88	7.50	8.77	13.14	15.18		
	HDD50	5760	1152	915	821	374	102	8	2	7	92	369	818	1100		
	HDD65	10278	1617	1335	1286	818	474	258	161	216	454	826	1268	1565		
	CDD50	993	0	0	0	6	94	205	324	267	88	9	0	0		
	CDD65	35	0	0	0	0	1	5	18	11	0	0	0	0		
	CDH74	676	0	0	0	2	37	95	285	206	50	1	0	0		
	CDH80	115	0	0	0	0	2	11	57	39	6	0	0	0		
	WSAvg	6.7	5.6	5.3	6.4	7.6	8.3	7.8	7.3	6.6	6.9	6.9	5.7	5.5		
	PrecAvg	19.7	1.5	1.0	1.0	0.9	1.7	3.0	2.7	2.7	2.0	1.0	1.2	1.3		
(15) Precipitation	PrecMax	31.5	5.0	3.1	2.0	2.2	6.3	7.2	6.9	6.3	5.1	2.3	3.1	5.1		
	PrecMin	12.8	0.2	0.1	0.1	0.1	0.1	0.6	0.5	0.6	0.1	0.1	0.0	0.0		
	PrecStd	4.3	1.1	0.7	0.6	0.7	1.3	2.0	1.7	1.5	1.2	0.6	0.9	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	44.8	51.5	68.1	78.5	80.9	85.9	84.6	80.1	68.0	53.4	46.4		
		MCWB	37.2	35.6	40.1	48.4	54.1	57.9	61.6	59.8	57.3	51.4	41.5	37.7		
	2%	DB	41.9	41.0	45.8	61.4	73.0	76.7	80.6	79.4	73.6	60.2	45.6	41.6		
		MCWB	35.2	34.0	36.7	45.4	52.0	56.5	59.7	59.7	55.5	47.2	38.5	35.2		
	5%	DB	39.0	38.3	42.6	56.2	69.0	72.9	76.8	75.4	68.4	55.0	41.3	38.1		
		MCWB	33.4	32.3	34.7	42.3	50.8	55.2	58.6	58.3	53.8	44.3	35.5	33.1		
	10%	DB	36.0	35.3	39.8	51.9	64.9	69.5	73.4	71.9	63.6	51.2	38.0	34.3		
		MCWB	31.3	30.1	32.9	39.8	48.7	53.7	57.4	56.8	51.2	42.0	33.2	30.1		
	0.4%	WB	37.9	36.7	41.1	49.6	57.9	60.8	64.9	64.4	59.8	52.8	42.8	38.4		
		MCDB	44.6	42.5	48.9	65.9	71.5	73.9	76.2	75.2	73.3	65.7	50.8	44.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.7	34.4	37.7	46.0	54.4	58.6	62.6	61.6	57.2	48.3	39.1	35.6		
		MCDB	41.1	40.3	44.7	60.1	68.4	72.3	75.2	74.1	69.8	57.9	44.5	40.9		
	5%	WB	33.9	32.5	35.2	43.1	52.0	56.8	60.7	59.7	54.6	45.3	36.1	33.0		
		MCDB	38.7	37.9	41.7	54.1	65.8	69.6	73.1	71.7	66.1	53.6	40.3	37.7		
	10%	WB	31.4	30.2	33.2	40.7	49.9	55.2	58.8	58.0	52.2	42.8	33.6	30.2		
		MCDB	35.5	35.3	39.3	50.7	62.9	66.4	69.9	69.3	62.7	50.1	37.7	34.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	15.6	16.3	18.4	21.2	20.3	20.9	21.5	20.3	16.3	13.2	14.0		
		MCDBR	16.1	16.2	18.4	26.4	28.1	26.1	27.4	28.4	29.0	23.4	17.0	15.5		
		MCWBR	12.1	11.4	12.0	14.4	13.2	11.2	11.4	12.1	13.9	13.4	12.0	12.2		
	5% WB	MCDBR	16.2	16.1	17.8	24.3	25.4	22.2	24.0	25.0	26.7	22.1	15.8	15.9		
		MCWBR	12.7	11.8	12.1	13.8	12.4	10.5	11.3	12.0	13.4	13.3	11.8	12.9		
(40) Clear-Sky Solar Irradiance	taub		0.235	0.250	0.297	0.322	0.347	0.346	0.345	0.346	0.308	0.286	0.246	0.231		
	taud		2.370	2.415	2.342	2.402	2.393	2.411	2.452	2.432	2.546	2.566	2.449	2.286		
	Ebn at Noon		236	274	279	287	284	285	283	276	275	257	231	206		
	Edh at Noon		15	21	30	32	35	34	32	31	24	19	14	13		
(44) All-Sky Solar Radiation	RadAvg		238	531	985	1466	1735	1898	1862	1521	1049	569	261	172		
	RadStd		22	40	76	88	123	154	125	128	99	60	22	14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page